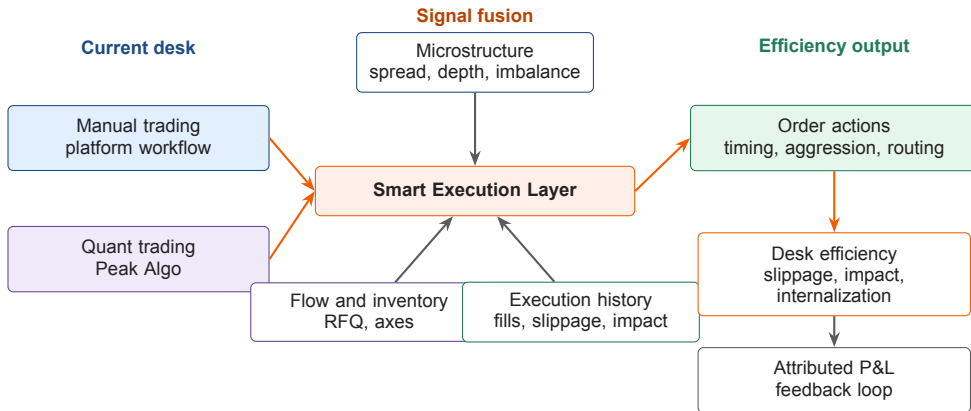


Applications of LOB and Microstructure

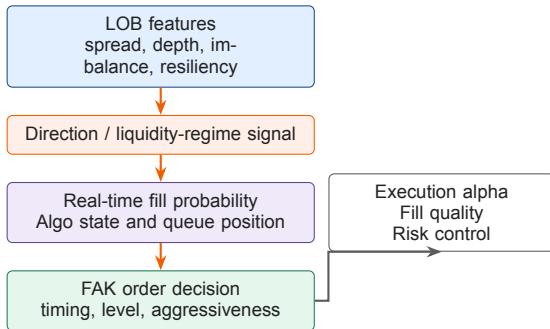
Algo Type1 Execution Alpha and Internal Flow Efficiency

Quant team | June 2026

■ Smart Execution as a Desk Efficiency Layer



■ FAK Order Execution Alpha



Case 1: favorable direction

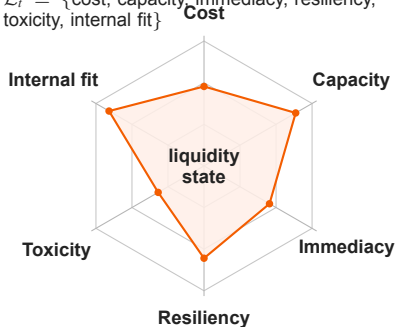
- ▶ Direction supports the order side.
- ▶ Expected fill probability is higher.
- ▶ Execution alpha: lower slippage and lower market impact.

Case 2: adverse direction

- ▶ Direction works against the FAK order.
- ▶ Revise fill expectation and aggressiveness.
- ▶ Improve success rate and reduce unhedged risk.

■ Liquidity is the True State Variable

$\mathcal{L}_t = \{\text{cost, capacity, immediacy, resiliency, toxicity, internal fit}\}$



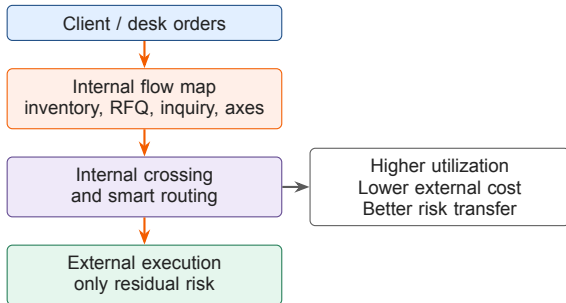
How to read the state

- ▶ **High capacity + fast resiliency:** raise FAK confidence.
- ▶ **High cost or toxicity:** revise fill expectation and aggression.
- ▶ **High internal fit:** cross internally before external routing.
- ▶ **Weak state:** delay, slice smaller, or hedge earlier.

Design rule

Price is the observation. Liquidity is the control state.

■ Microstructure for Internal Flow



Evidence-backed plan

- ▶ Segment flow by liquidity regime before routing.
- ▶ Prioritize internal crossing before paying spread and impact.
- ▶ Use queue and order-flow signals for residual external execution.
- ▶ Validate by internalization uplift and residual external cost.

■ How We Save 0.01 bp Per Trade

Base-case value

90M

annual P&L target from 0.01 bp saving

- ▶ 2025 flow: 45tn.
- ▶ Average duration: 2 years.
- ▶ $45tn \times 2 \times 0.01bp \approx 90M$.

Where the saving comes from

- ▶ **Internal crossing:** avoid spread and external market impact.
- ▶ **Timing:** trade when depth and resiliency are favorable.
- ▶ **Adverse-selection control:** avoid passive fills when direction is unfavorable.

How we prove it

- ▶ Slippage vs. arrival / decision price.
- ▶ Market impact after execution.
- ▶ Internalization uplift and residual external cost.